# Notes on Gauge Theory

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# Appendix

## Vector valued-differential forms

Through out this section, let $M$ be a smooth manifold and $V$ be a vector space. $\Omega^k(M)$ denotes the space of smooth $k$-forms on $M$.

### $V$-valued forms

**Definition 0.1.** Define the space of **$V$-valued $k$-forms** on $M$ as

$$\Omega^k(M,V) := C^\infty(M,(\wedge^k T^*M)\otimes V).$$

Fix a basis $\{v_i\}$ of $V$, then any $\alpha \in \Omega^k(M,V)$ can be written as

$$\alpha = \alpha^i \otimes v_i (= \alpha^i v_i), \quad \alpha^i \in \Omega^k(M).$$

**Scalar multiplication:** We can define a scalar multiplication

$$\begin{aligned} \Omega^a(M) \times \Omega^k(M,V) &\to \Omega^{a+k}(M,V) \\ (\lambda, \alpha\otimes v) &\mapsto (\lambda\wedge\alpha)\otimes v. \end{aligned}$$

If $\alpha = \alpha^i v_i$ then $\lambda\cdot\alpha := (\lambda\wedge\alpha^i)\otimes v_i$. More generally, we can define

$$\begin{aligned} \Omega^a(M,\mathrm{End}(V)) \times \Omega^k(M,V) &\to \Omega^{a+k}(M,V) \\ (\eta\otimes A, \alpha\otimes v) &\mapsto (\eta\wedge\alpha)\otimes Av. \end{aligned}$$

**Exterior derivative:** For $\alpha\in\Omega^k(M,V)$, define

$$\mathrm{d}\alpha := (\mathrm{d}\alpha^i)\otimes v_i \in \Omega^{k+1}(M,V).$$

Still, this definition is well-defined.

**Proposition 0.2.** *The exterior derivative* d *satisfies the following properties:*

- $\mathrm{d}^2 = 0$.
- $\mathrm{d}(\lambda\cdot\alpha) = \mathrm{d}\lambda\cdot\alpha + (-1)^a\lambda\cdot\mathrm{d}\alpha,\ \forall\,\lambda\in\Omega^a(M), \alpha\in\Omega^k(M,V),$

### Products of vector-valued forms

Let $V, W, Z$ be vector spaces and $\mu : V\times W \to Z$ be a bilinear map.

**Definition 0.3.** We can define a product:

$$\begin{aligned} \Omega^k(M,V)\times\Omega^l(M,W) &\to \Omega^{k+l}(M,Z) \\ (\alpha,\beta) &\mapsto \alpha\cdot\beta \end{aligned}$$

where

$$\begin{aligned} (\alpha\cdot\beta)_p(X_1,\cdots,X_{k+l}) &:= \frac{1}{k!l!}\sum_{\sigma\in S_{k+l}} \mathrm{sgn}(\sigma)\mu\Big(\alpha_p(X_{\sigma(1)},\cdots,X_{\sigma(k)}), \beta_p(X_{\sigma(k+1)},\cdots,X_{\sigma(k+l)})\Big) \\ &= \sum_{(k,l)\text{-shuffles }\sigma} \mathrm{sgn}(\sigma)\mu\Big(\alpha_p(X_{\sigma(1)},\cdots,X_{\sigma(k)}), \beta_p(X_{\sigma(k+1)},\cdots,X_{\sigma(k+l)})\Big).^1 \end{aligned}$$

[1] We say $\sigma\in S_{k+l}$ is a $(k,l)$-shuffle if $\sigma(1)<\cdots<\sigma(k)$ and $\sigma(k+1)<\cdots<\sigma(k+l)$.

**Proposition 0.4.** *This product satisfies the following properties:*

- *If* $\alpha = \alpha^i v_i$ *and* $\beta = \beta^j w_j$*, then* $\alpha \cdot \beta = (\alpha^i \wedge \beta^j) \otimes \mu(v_i, w_j)$.
- *If* $\lambda \in \Omega^a(M)$*, then* $\lambda \cdot (\alpha \cdot \beta) = (\lambda \cdot \alpha) \cdot \beta = (-1)^{ka} \alpha \cdot (\lambda \cdot \beta)$.
- $\mathrm{d}(\alpha \cdot \beta) = \mathrm{d}\alpha \cdot \beta + (-1)^k \alpha \cdot \mathrm{d}\beta$.

Notice that in general, $\alpha \cdot \beta$ is not associative, commutative, or anticommutative.

**Example 0.5.** Let $\mu : \mathbb{R}^{m\times n} \times \mathbb{R}^{n \times p} \to \mathbb{R}^{m \times p}$ be the matrix multiplication. $\alpha = \alpha^{ij} e_{ij} \in \Omega^k(M, \mathbb{R}^{m\times n})$ and $\beta = \beta^{ij} e_{ij} \in \Omega^l(M, \mathbb{R}^{n\times p})$, then

$$\alpha \cdot \beta = (\alpha^{ij} \wedge \beta^{jk}) e_{ik} =: \alpha \wedge \beta$$

is the usual matrix-valued wedge product.

### $\mathfrak{g}$-valued forms

In the special case when $V = \mathfrak{g}$ is a Lie algebra or more precisely the matrix algebra $\mathfrak{gl}(n, \mathbb{R})$, we take $\mu : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ to be the Lie bracket, and we write $[\alpha, \beta]$ instead of $\alpha \cdot \beta$.

$$\begin{aligned}[\alpha,\beta]_p(X_1, \cdots, X_{k+l}) &:= \frac{1}{k!l!} \sum_{\sigma \in S_{k+l}} \operatorname{sgn}(\sigma) [\alpha_p(X_{\sigma(1)}, \cdots, X_{\sigma(k)}), \beta_p(X_{\sigma(k+1)}, \cdots, X_{\sigma(k+l)})] \\ &= \sum_{(k,l)\text{-shuffles } \sigma} \operatorname{sgn}(\sigma) [\alpha_p(X_{\sigma(1)}, \cdots, X_{\sigma(k)}), \beta_p(X_{\sigma(k+1)}, \cdots, X_{\sigma(k+l)})].\end{aligned}$$

For example, if $\alpha, \beta \in \Omega^1(M, \mathfrak{g})$, then $[\alpha, \beta]_p(X, Y) = [\alpha_p(X), \beta_p(Y)] - [\alpha_p(Y), \beta_p(X)]$.

**Example 0.6.** If $\mathfrak{g} = \mathfrak{gl}(n, \mathbb{R})$, $[A, B] = AB - BA$, then

$$[\alpha, \beta] = \alpha \wedge \beta - (-1)^{|\alpha||\beta|} \beta \wedge \alpha.$$

Hence

$$[\alpha, \alpha] = \begin{cases} 2\alpha \wedge \alpha, & \text{if } |\alpha| \text{ is odd,} \\ 0, & \text{if } |\alpha| \text{ is even.} \end{cases}$$

**Proposition 0.7.** *For* $\alpha = \alpha^i A_i, \beta = \beta^j A_j$*, we have*

$$[\alpha, \beta] = (\alpha^i \wedge \beta^j) \otimes [A_i, A_j].$$

**Corollary 0.8.** *If* $\alpha \in \Omega^k(M, \mathfrak{g})$ *and* $\beta \in \Omega^l(M, \mathfrak{g})$*, then*

$$[\alpha, \beta] = (-1)^{kl} [\beta, \alpha].$$

**Example 0.9.** If $\alpha, \beta \in \Omega^1(M, \mathfrak{g})$, then $[\alpha, \beta] = [\beta, \alpha]$, thus $[\alpha, \alpha]$ is not necessarily zero.

**Proposition 0.10.** *For* $\alpha \in \Omega^k(M, \mathfrak{g})$*,* $\beta \in \Omega^l(M, \mathfrak{g})$ *and* $\gamma \in \Omega^m(M, \mathfrak{g})$*, we have*

$$[\alpha, [\beta, \gamma]] + (-1)^{k(l+m)} [\beta, [\gamma, \alpha]] + (-1)^{m(k+l)} [\gamma, [\alpha, \beta]] = 0.$$

**Example 0.11.** If $\alpha \in \Omega^1(M, \mathfrak{g})$, then $[\alpha, [\alpha, \alpha]] = 0$.

**Proposition 0.12.** $\mathrm{d}[\alpha, \beta] = [\mathrm{d}\alpha, \beta] + (-1)^{|\alpha|} [\alpha, \mathrm{d}\beta]$.

### Vector Bundle valued forms

Let $E \to M$ be a vector bundle. We can define

$$\Omega^k(M,E) := C^\infty(M, (\wedge^k T^*M) \otimes E).$$

Let $e_1, \cdots, e_n$ be a local frame of $E$ over an open subset $U \subset M$, then any $\alpha \in \Omega^k(M,E)$ can be written as

$$\alpha = \alpha^i e_i (= \alpha^i \otimes e_i), \quad \alpha^i \in \Omega^k(U).$$

**Example 0.13.** Let $\nabla$ be a connection on $E$, then $\nabla^2$ is a $\mathrm{End}(E)$-valued 2-form on $M$.

### Maurer-Cartan forms

Let $G$ be a Lie group and $\mathfrak{g}$ be its Lie algebra.

**Definition 0.14.** The **Maurer-Cartan form** $\theta$ on $G$ is the unique $\mathfrak{g}$-valued 1-form $\theta \in \Omega^1(G, \mathfrak{g})$ such that

- $\theta_e(X_e) = X_e,\ \forall X_e \in T_eG = \mathfrak{g}$.
- $\theta$ is left-invariant, i.e. $(l_g)^*\theta = \theta,\ \forall g \in G$. That is

$$\theta_g(X_g) = (l^*_{g^{-1}}\theta_e)(X_g) = \theta_e(l_{g^{-1}*}X_g) = l_{g^{-1}*}X_g, \quad \forall X_g \in T_gG.$$

**Proposition 0.15.** *The Maurer-Cartan form $\theta$ satisfies the* ***Maurer-Cartan equation***

$$\mathrm{d}\theta + \frac{1}{2}[\theta,\theta] = 0.$$

**Proof.** It suffices to check this equation on left-invariant vector fields. For any left-invariant vector field $X$, $\theta(X)$ is a constant function on $G$, thus

$$(\mathrm{d}\theta)(X,Y) = X(\theta(Y)) - Y(\theta(X)) - \theta([X,Y]) = -[X_e, Y_e] \text{ (constant } \mathfrak{g}\text{-valued function)}$$
$$\frac{1}{2}[\theta,\theta](X,Y) = \frac{1}{2}([\theta(X),\theta(Y)] - [\theta(Y),\theta(X)]) = [X_e, Y_e]. \text{ (constant } \mathfrak{g}\text{-valued function)}$$

Thus

$$(\mathrm{d}\theta + \frac{1}{2}[\theta,\theta])(X,Y) = 0.$$

□

**Proposition 0.16.** *Under right translation, the Maurer-Cartan form $\theta$ transforms as*

$$(r_g)^*\theta = (\mathrm{Ad}_{g^{-1}})\theta, \quad \forall g \in G.$$

**Proof.** It suffices to check this equation on left-invariant vector fields. For any left-invariant vector field $X$, both $(r_g^*\theta)(X)$ and $\mathrm{Ad}_{g^{-1}}(\theta(X))$ are constant $\mathfrak{g}$-valued functions on $G$, and

$$(r_g^*\theta_g)(X_e) = \theta_g(r_{g*}X_e) = l_{g^{-1}*}(r_{g*}X_e) = \mathrm{Ad}_{g^{-1}}(X_e) = \mathrm{Ad}_{g^{-1}}(\theta_e(X_e)).$$

□